

Actualizar Ubuntu.

```
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# apt update
Hit:1 https://deb.nodesource.com/node_22.x nodistro InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 https://ppa.launchpadcontent.net/gns3/ppa/ubuntu noble InRelease [18.0 kB]
E: Repository 'https://ppa.launchpadcontent.net/gns3/ppa/ubuntu noble InRelease' changed its 'Label' value from 'GNS3 PP
A' to 'GNS3 v2 PPA'
N: This must be accepted explicitly before updates for this repository can be applied. See apt-secure(8) manpage for det
ails.
Do you want to accept these changes and continue updating from this repository? [y/N] y
Get:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1066 kB]
Get:8 https://ppa.launchpadcontent.net/gns3/ppa/ubuntu noble/main amd64 Packages [2352 B]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [220 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1337 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1216 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [301 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1699 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1566 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [361 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [341 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1689 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [386 kB]
```

```
root@BOG30487:/mnt/c/Users/jbara# apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libllvm19 libqt5core5t64 libqt5dbus5t64 libqt5designer5 libqt5gui5t64 libqt5help5 libqt5network5t64
  libqt5printsupport5t64 libqt5qml5 libqt5qmlmodels5 libqt5quick5 libqt5sql5-sqlite libqt5sql5t64 libqt5svg5
  libqt5test5t64 libqt5waylandclient5 libqt5waylandcompositor5 libqt5websockets5 libqt5widgets5t64 libqt5xml5t64
  libxcb-xinerama0 libxcb-xinput0 python3-pyqt5 python3-qtsvg python3-pyqt5.qtwebsockets python3-pyqt5.sip
  qt5-gtk-platformtheme qttranslations5-l10n qtwayland5
Use 'sudo apt autoremove' to remove them.
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  libzvb1-common libqt5xml5t64 libsrt1.5-gnutls qt5-gtk-platformtheme
  libcommons-io-java libcommons-beanutils-java libqt5sql5t64 libqt5qml5
  libqt5test5t64 python3-pip-whl libbson1 libqt5quick5 python3-wheel
  libqt5gui5t64 libmbdecrypto7t64 libavcodec60 libzvb10t64
  libqt5printsupport5t64 libcommons-lang3-java libavutil58 libqt5widgets5t64
  libswscale7 python3-pip libqt5dbus5t64 libqt5network5t64 libswresample4
  libqt5sql5-sqlite libqt5qmlmodels5 libavformat60 libqt5core5t64
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following NEW packages will be installed:
  libqt6openglwidgets6t64 libqt6sql6-sqlite libqt6sql6t64 libqt6svgwidgets6 libqt6test6t64 libqt6websockets6
  libqt6xml6t64 python3-pyqt6 python3-pyqt6.qtsvg python3-pyqt6.qtwebsockets python3-pyqt6.sip
The following upgrades have been deferred due to phasing:
  dnsmasq-base libegl-mesa0 libgbm1 libgl1-mesa-dri libglx-mesa0 mesa-libgallium mesa-va-drivers mesa-udpau-drivers
  mesa-vulkan-drivers
The following packages will be upgraded:
  alsa-ucm-conf apparmor apport apport-core-dump-handler base-files binutils binutils-common binutils-x86-64-linux-gnu
  byobu cloud-init console-setup console-setup-linux dconf-gsettings-backend dconf-service dhcpcd-base dmidecode
```

Instalar herramientas base.

```
root@BOG30487:/mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# apt install -y curl wget git pciutils lshw htop net-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.15).
wget is already the newest version (1.21.4-1ubuntu4.5).
wget set to manually installed.
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
lshw is already the newest version (02.19.git.2021.06.19.996aaad9c7-2ubuntu0.24.04.1).
lshw set to manually installed.
net-tools is already the newest version (2.10-0.1ubuntu4.4).
The following packages were automatically installed and are no longer required:
libllvm19 libqt5core5t64 libqt5dbus5t64 libqt5designer5 libqt5gui5t64 libqt5help5 libqt5network5t64
libqt5printsupport5t64 libqt5qml5 libqt5qmlmodels5 libqt5quick5 libqt5sql5-sqlite libqt5sql5t64 libqt5svg5
libqt5test5t64 libqt5waylandclient5 libqt5waylandcompositor5 libqt5websockets5 libqt5widgets5t64 libqt5xml5t64
libxcb-xinerama0 libxcb-xinput0 python3-pyqt5 python3-pyqt5.qtsvg python3-pyqt5.qtwebsockets python3-pyqt5.sip
qt5-gtk-platformtheme qttranslations5-l10n qtwayland5
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
libpci3
Suggested packages:
lm-sensors strace
The following NEW packages will be installed:
htop libpci3 pciutils
0 upgraded, 3 newly installed, 0 to remove and 9 not upgraded.
Need to get 277 kB of archives.
After this operation, 735 kB of additional disk space will be used.
```

Instalar Ollama

```
root@BOG30487:/mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# curl -fsSL https://ollama.com/install.sh | sh
>>> Cleaning up old version at /usr/local/lib/ollama
>>> Installing ollama to /usr/local
>>> Downloading ollama-linux-amd64.tar.zst
##### 100.0%
>>> Creating ollama user...
>>> Adding ollama user to render group...
>>> Adding ollama user to video group...
>>> Adding current user to ollama group...
>>> Creating ollama systemd service...
>>> Enabling and starting ollama service...
Created symlink /etc/systemd/system/default.target.wants/ollama.service → /etc/systemd/system/ollama.service.
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.
root@BOG30487:/mnt/c/Users/jbara#
```

Verificar la instalación.

```
root@BOG30487: /mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# ollama -v
ollama version is 0.35.0
root@BOG30487:/mnt/c/Users/jbara# which ollama
/usr/local/bin/ollama
root@BOG30487:/mnt/c/Users/jbara#
```

Comprobar el servicio.

```
root@BOG30487: /mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# systemctl status ollama --no-pager
● ollama.service - Ollama Service
   Loaded: loaded (/etc/systemd/system/ollama.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-30 10:48:26 -05; 1min 38s ago
     Main PID: 6514 (ollama)
        Tasks: 10 (limit: 7100)
      Memory: 91.2M (peak: 113.1M)
         CPU: 517ms
       CGroup: /system.slice/ollama.service
               └─6514 /usr/local/bin/ollama serve

Sep 30 10:48:26 BOG30487 ollama[6514]: time=2026-09-30T10:48:26.998-05:00 level=INFO source=routes.go:2099 msg="server ...
Sep 30 10:48:26 BOG30487 ollama[6514]: time=2026-09-30T10:48:26.998-05:00 level=INFO source=routes.go:2101 msg="_: false"
Sep 30 10:48:26 BOG30487 ollama[6514]: time=2026-09-30T10:48:26.999-05:00 level=INFO source=images.go:922 msg="...lobs: 0"
Sep 30 10:48:26 BOG30487 ollama[6514]: time=2026-09-30T10:48:26.999-05:00 level=INFO source=images.go:930 msg="...oved: 0"
Sep 30 10:48:27 BOG30487 ollama[6514]: time=2026-09-30T10:48:27.000-05:00 level=INFO source=routes.go:2159 msg="...0.35.0)"
Sep 30 10:48:27 BOG30487 ollama[6514]: time=2026-09-30T10:48:27.001-05:00 level=INFO source=runner.go:60 msg="d...GPUs..."
Sep 30 10:48:27 BOG30487 ollama[6514]: time=2026-09-30T10:48:27.288-05:00 level=INFO source=model_recommendatio...ilures=0
Sep 30 10:48:27 BOG30487 ollama[6514]: time=2026-09-30T10:48:27.732-05:00 level=INFO source=types.go:50 msg="in...5.2 GiB"
Sep 30 10:48:27 BOG30487 ollama[6514]: time=2026-09-30T10:48:27.732-05:00 level=INFO source=routes.go:2209 msg="...ctx=4096
Sep 30 10:48:59 BOG30487 ollama[6514]: [GIN] 2026/09/30 - 10:48:59 | 200 | 195.451µs | 127.0.0.1 | GE_version"
Hint: Some lines were ellipsized, use -l to show in full.
root@BOG30487:/mnt/c/Users/jbara#
```

Habilitar e iniciar Ollama.

```
root@BOG30487:/mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# systemctl enable --now ollama
root@BOG30487:/mnt/c/Users/jbara# systemctl status ollama --no-pager
● ollama.service - Ollama Service
   Loaded: loaded (/etc/systemd/system/ollama.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-30 10:48:26 -05; 3min 12s ago
     Main PID: 6514 (ollama)
       Tasks: 10 (Limit: 7100)
      Memory: 91.2M (peak: 113.1M)
         CPU: 520ms
        CGroup: /system.slice/ollama.service
                └─6514 /usr/local/bin/ollama serve

Sep 30 10:48:26 BOG30487 ollama[6514]: time=2026-09-30T10:48:26.998-05:00 level=INFO source=routes.go:2099 msg="server ...
Sep 30 10:48:26 BOG30487 ollama[6514]: time=2026-09-30T10:48:26.998-05:00 level=INFO source=routes.go:2101 msg="...: false"
Sep 30 10:48:26 BOG30487 ollama[6514]: time=2026-09-30T10:48:26.999-05:00 level=INFO source=images.go:922 msg="...lobs: 0"
Sep 30 10:48:26 BOG30487 ollama[6514]: time=2026-09-30T10:48:26.999-05:00 level=INFO source=images.go:930 msg="...oved: 0"
Sep 30 10:48:27 BOG30487 ollama[6514]: time=2026-09-30T10:48:27.000-05:00 level=INFO source=routes.go:2159 msg="...0.35.0)"
Sep 30 10:48:27 BOG30487 ollama[6514]: time=2026-09-30T10:48:27.001-05:00 level=INFO source=runner.go:60 msg="d_GPUs..."
Sep 30 10:48:27 BOG30487 ollama[6514]: time=2026-09-30T10:48:27.288-05:00 level=INFO source=model_recommendatio...ilures=0
Sep 30 10:48:27 BOG30487 ollama[6514]: time=2026-09-30T10:48:27.732-05:00 level=INFO source=types.go:50 msg="in...5.2 GiB"
Sep 30 10:48:27 BOG30487 ollama[6514]: time=2026-09-30T10:48:27.732-05:00 level=INFO source=routes.go:2209 msg="...ctx=4096
Sep 30 10:48:59 BOG30487 ollama[6514]: [GIN] 2026/09/30 - 10:48:59 | 200 | 195.451µs | 127.0.0.1 | GE...version"
Hint: Some lines were ellipsized, use -l to show in full.
root@BOG30487:/mnt/c/Users/jbara#
```

Comprobar la API local

```
root@BOG30487:/mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# curl http://127.0.0.1:11434/api/tags
{"models": []}root@BOG30487:/mnt/c/Users/jbara#
```

Descarga, ejecución y administración de modelos.

```
root@BOG30487:/mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# ollama pull llama3.2:1b
pulling manifest
pulling 74701a8c35f6: 100% | 1.3 GB/1.3 GB 9.6 MB/s 0s
verifying sha256 digest
writing manifest
success
root@BOG30487:/mnt/c/Users/jbara# ollama run llama3.2:1b
>>>
Use Ctrl + d or /bye to exit.
>>>
root@BOG30487:/mnt/c/Users/jbara# ollama list
NAME ID SIZE MODIFIED
llama3.2:1b baf6a787fdff 1.3 GB 56 seconds ago
root@BOG30487:/mnt/c/Users/jbara# ollama ps
NAME ID SIZE PROCESSOR CONTEXT UNTIL
llama3.2:1b baf6a787fdff 1.5 GB 100% CPU 4096 4 minutes from now
root@BOG30487:/mnt/c/Users/jbara# ollama show llama3.2:1b
Model
architecture llama
parameters 1.2B
context length 131072
embedding length 2048
quantization Q8_0

Capabilities
completion
tools
```

```
root@BOG30487:/mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# ollama stop llama3.2:1b
root@BOG30487:/mnt/c/Users/jbara# ollama rm llama3.2:1b
deleted 'llama3.2:1b'
root@BOG30487:/mnt/c/Users/jbara# ollama cp llama3.2:1b Asistente
Error: model "llama3.2:1b" not found
root@BOG30487:/mnt/c/Users/jbara# ollama pull llama3.2:1b
pulling manifest
pulling 74701a8c35f6: 100% | 1.3 GB/1.3 GB 7.4 MB/s 0s
pulling fcc5a6bec9da: 100% | 7.7 KB
pulling a70ff7e570d9: 100% | 6.0 KB
pulling 4f659a1e86d7: 100% | 485 B
verifying sha256 digest
writing manifest
success
root@BOG30487:/mnt/c/Users/jbara# ollama cp llama3.2:1b Asistente
copied 'llama3.2:1b' to 'Asistente'
root@BOG30487:/mnt/c/Users/jbara#
```

Construcción de un modelo personalizado.


```
root@BOG30487: /mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.5 LTS
Release:        24.04
Codename:       noble
root@BOG30487:/mnt/c/Users/jbara# uname -a
Linux BOG30487 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64
GNU/Linux
root@BOG30487:/mnt/c/Users/jbara# lscpu
Architecture:        x86_64
CPU op-mode(s):      32-bit, 64-bit
Address sizes:        39 bits physical, 48 bits virtual
Byte Order:           Little Endian
CPU(s):               4
On-line CPU(s) list: 0-3
Vendor ID:            GenuineIntel
Model name:           13th Gen Intel(R) Core(TM) i7-1355U
CPU family:           6
Model:                186
Thread(s) per core:   2
Core(s) per socket:   2
Socket(s):            1
Stepping:              3
BogoMIPS:             5222.39
Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse s
se2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopt xtopology tsc_reliable nonst
op_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
```

```
root@BOG30487: /mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# free -h
              total        used        free      shared  buff/cache   available
Mem:           5.8Gi       523Mi       5.2Gi       3.8Mi       248Mi       5.3Gi
Swap:          2.0Gi          0B        2.0Gi
root@BOG30487:/mnt/c/Users/jbara# lsblk
NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda  8:0  0 388.4M  1 disk
sdb  8:16 0  186M  1 disk
sdc  8:32 0    2G  0 disk [SWAP]
sdd  8:48 0    1T  0 disk /mnt/wslg/distro
/
root@BOG30487:/mnt/c/Users/jbara# df -h
Filesystem      Size  Used Avail Use% Mounted on
none            2.9G   0 2.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            2.9G  4.0K 2.9G   1% /mnt/wsl
drivers         476G  280G  196G  59% /usr/lib/wsl/drivers
/dev/sdd        1007G   11G  945G   2% /
none            2.9G 104K 2.9G   1% /mnt/wslg
none            2.9G   0 2.9G   0% /usr/lib/wsl/lib
rootfs          2.9G  2.7M 2.9G   1% /init
none            2.9G  548K 2.9G   1% /run
none            2.9G   0 2.9G   0% /run/lock
none            2.9G   0 2.9G   0% /run/shm
none            2.9G   76K 2.9G   1% /mnt/wslg/versions.txt
none            2.9G   76K 2.9G   1% /mnt/wslg/doc
C:\             476G  280G  196G  59% /mnt/c
tmpfs           2.9G  220K 2.9G   1% /run/qemu
```

Ejercicio 2 – Gestión de paquetes.

```
root@BOG30487:/mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 https://deb.nodesource.com/node_22.x nodistro InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:6 https://ppa.launchpadcontent.net/gns3/ppa/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@BOG30487:/mnt/c/Users/jbara# apt install -y htop curl git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
htop is already the newest version (3.3.0-4build1).
curl is already the newest version (8.5.0-2ubuntu10.15).
git is already the newest version (1:2.43.0-1ubuntu7.3).
The following packages were automatically installed and are no longer required:
 libllvm19 libqt5core5t64 libqt5dbus5t64 libqt5designer5 libqt5gui5t64 libqt5help5 libqt5network5t64
 libqt5printsupport5t64 libqt5qml5 libqt5qmlmodels5 libqt5quick5 libqt5sql5-sqlite libqt5sql5t64 libqt5svg5
 libqt5test5t64 libqt5waylandclient5 libqt5waylandcompositor5 libqt5websockets5 libqt5widgets5t64 libqt5xml5t64
 libxcb-xinerama0 libxcb-xinput0 python3-pyqt5 python3-pyqt5.qtsvg python3-pyqt5.qtwebsockets python3-pyqt5.sip
 qt5-gtk-platformtheme qttranslations5-l10n qtwayland5
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 9 not upgraded.
root@BOG30487:/mnt/c/Users/jbara#
```

```
root@BOG30487:/mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# apt policy htop curl git
htop:
  Installed: 3.3.0-4build1
  Candidate: 3.3.0-4build1
  Version table:
 *** 3.3.0-4build1 500
    500 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages
    100 /var/lib/dpkg/status
curl:
  Installed: 8.5.0-2ubuntu10.15
  Candidate: 8.5.0-2ubuntu10.15
  Version table:
 *** 8.5.0-2ubuntu10.15 500
    500 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages
    500 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages
    100 /var/lib/dpkg/status
 8.5.0-2ubuntu10 500
    500 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages
git:
  Installed: 1:2.43.0-1ubuntu7.3
  Candidate: 1:2.43.0-1ubuntu7.3
  Version table:
 *** 1:2.43.0-1ubuntu7.3 500
    500 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages
    500 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages
    100 /var/lib/dpkg/status
 1:2.43.0-1ubuntu7 500
    500 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages
```

Ejercicio 3 – Procesos antes y después de la IA

```
root@BOG30487: /mnt/c/Use  X + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# ps aux --sort=-%cpu | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
ollama    6514  0.4  0.6 2614576 36544 ?        Ssl  12:14   1:34 /usr/local/bin/ollama serve
root      7868  0.0  0.3 370108 20096 ?        Ssl  17:34   0:00 /usr/libexec/packagekitd
root       1  0.0  0.2 22480 13160 ?        Ss   12:04   0:04 /sbin/init
root      621  0.0  0.1 14328  7000 pts/0    S+   12:04   0:01 sudo su
message+  166  0.0  0.0 10064  5120 ?        Ss   12:04   0:01 @dbus-daemon --system --address=systemd: --nofork --n
opidfile --systemd-activation --syslog-only
root      457  0.0  0.0  3080   896 ?        S    12:04   0:00 /init
root      42  0.0  0.2 66900 15612 ?        S<s  12:04   0:00 /usr/lib/systemd/systemd-journald
polkitd   991  0.0  0.1 308180 7680 ?        Ssl  12:05   0:00 /usr/lib/polkit-1/polkitd --no-debug
root      182  0.0  0.1 18168  8448 ?        Ss   12:04   0:00 /usr/lib/systemd/systemd-logind
root@BOG30487:/mnt/c/Users/jbara# ps aux --sort=-%mem | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
ollama    6514  0.4  0.6 2614576 36544 ?        Ssl  12:14   1:34 /usr/local/bin/ollama serve
root      247  0.0  0.3 107028 22268 ?        Ssl  12:04   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unatt
ended-upgrade-shutdown --wait-for-signal
root      7868  0.0  0.3 370108 20096 ?        Ssl  17:34   0:00 /usr/libexec/packagekitd
root      42  0.0  0.2 66900 15612 ?        S<s  12:04   0:00 /usr/lib/systemd/systemd-journald
root     191  0.0  0.2 68744 14720 ?        Ss   12:04   0:00 /usr/sbin/virtlogd
root     188  0.0  0.2 68740 14592 ?        Ss   12:04   0:00 /usr/sbin/virtlockd
root       1  0.0  0.2 22480 13160 ?        Ss   12:04   0:04 /sbin/init
systemd+  154  0.0  0.2 21344 12672 ?        Ss   12:04   0:00 /usr/lib/systemd/systemd-resolved
root      626  0.0  0.1 20328 11264 ?        Ss   12:04   0:00 /usr/lib/systemd/systemd --user
root@BOG30487:/mnt/c/Users/jbara#
```

```
root@BOG30487: /mnt/c/Use  X root@BOG30487: /mnt/c/User  X + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# ps aux --sort=-%cpu | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
ollama    8060 23.4 24.7 2136124 1504460 ?        Rl   17:38   0:12 /usr/local/lib/ollama/llama-server --model /usr/share
/ollama/.ollama/models/blobs/sha256-74701a8c35f6c8d9a4b91f3f3497643001d63e0c7a84e085bed452548fa88d45 --port 37815 --host
127.0.0.1 --no-webui --offline -c 4096 -np 1 --log-verbosity 4 --no-log-prefix --no-log-timestamps --no-jinja --chat-te
mplate chatml --load-mode none --flash-attn auto -b 512 -ub 512 --context-shift --keep 4
ollama    6514  0.4  0.6 2614576 39692 ?        Ssl  12:14   1:34 /usr/local/bin/ollama serve
root     8053  0.2  0.4 1737260 29056 pts/4    Sl+  17:38   0:00 ollama run Asistente
root     8025  0.1  0.1 14332  6996 pts/3     S+   17:37   0:00 sudo su
root     8005  0.0  0.0  3080  1024 ?        S    17:37   0:00 /init
root     8028  0.0  0.0  5812  4736 pts/4     S    17:37   0:00 bash
root       1  0.0  0.2 22480 13160 ?        Ss   12:04   0:04 /sbin/init
jbara    8012  0.0  0.0  6076  5248 pts/3     Ss   17:37   0:00 -bash
root     7868  0.0  0.3 370108 20096 ?        Ssl  17:34   0:00 /usr/libexec/packagekitd
root@BOG30487:/mnt/c/Users/jbara# ps aux --sort=-%mem | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
ollama    8060 32.2 24.7 2136124 1504588 ?        Rl   17:38   0:17 /usr/local/lib/ollama/llama-server --model /usr/share
/ollama/.ollama/models/blobs/sha256-74701a8c35f6c8d9a4b91f3f3497643001d63e0c7a84e085bed452548fa88d45 --port 37815 --host
127.0.0.1 --no-webui --offline -c 4096 -np 1 --log-verbosity 4 --no-log-prefix --no-log-timestamps --no-jinja --chat-te
mplate chatml --load-mode none --flash-attn auto -b 512 -ub 512 --context-shift --keep 4
ollama    6514  0.4  0.6 2614576 39692 ?        Ssl  12:14   1:34 /usr/local/bin/ollama serve
root     8053  0.2  0.4 1737260 29056 pts/4    Sl+  17:38   0:00 ollama run Asistente
root     247  0.0  0.3 107028 22268 ?        Ssl  12:04   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unatt
ended-upgrade-shutdown --wait-for-signal
root     7868  0.0  0.3 370108 20096 ?        Ssl  17:34   0:00 /usr/libexec/packagekitd
root      42  0.0  0.2 66900 15612 ?        S<s  12:04   0:00 /usr/lib/systemd/systemd-journald
root     191  0.0  0.2 68744 14720 ?        Ss   12:04   0:00 /usr/sbin/virtlogd
root     188  0.0  0.2 68740 14592 ?        Ss   12:04   0:00 /usr/sbin/virtlockd
```

```
root@BOG30487:/mnt/c/User x root@BOG30487:/mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# ollama run Asistente
>>> Dime el proceso de mejora de una vpn gateway y el trafico de un tunel vpn para evitar la perdida de paquetes o filtr
... acción de la información sensible
¡Claro! A continuación, te proporciono un resumen del proceso de mejora de una VPN gateway y el tráfico de un
túnel VPN para evitar la pérdida de paquetes o filtración de la información sensible:

**Mejora de la VPN Gateway**

La mejora de la VPN gateway implica mejorar la calidad del tráfico y la seguridad de la conexión a Internet. A
continuación, se presentan los pasos generales para mejorar la VPN gateway:

1. **Verificación y actualización de software**: Asegúrate de que el software de la VPN gateway esté actualizado y
corregido cualquier bug o incompatibilidad.
2. **Optimización del rediseño**: Ajusta el rediseño de la VPN gateway para mejorar la velocidad y la eficiencia
de la conexión.
3. **Configuración de las políticas de seguridad**: Ajusta las políticas de seguridad para limitar el tráfico y la
acceso a Internet.
4. **Implementación de la autenticación multi-factor**: Implementa la autenticación multi-factor para agregar una
capa adicional de seguridad.
5. **Uso de la autenticación con protocolo HTTPS**: Utiliza la autenticación con protocolo HTTPS para proteger la
información de la conexión.

**Tráfico de un túnel VPN**

Para evitar la pérdida de paquetes o filtración de la información sensible, se utilizan técnicas de manejo del
tráfico en túneles VPN. A continuación, se presentan algunas de las técnicas más comunes:

1. **Tuning del bitrate**: Ajusta el bitrate del tráfico para minimizar la pérdida de paquetes y garantizar una
```

```
root@BOG30487:/mnt/c/User x root@BOG30487:/mnt/c/User x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# ps aux --sort=-%cpu | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
ollama    8060  63.8  24.7 2136124 1504716 ?        Sl   17:38   1:34 /usr/local/lib/ollama/llama-server --model /usr/share
/ollama/.ollama/models/blobs/sha256-74701a8c35f6c8d9a4b91f3f3497643001d63e0c7a84e085bed452548fa88d45 --port 37815 --host
127.0.0.1 --no-webui --offline -c 4096 -np 1 --log-verbosity 4 --no-log-prefix --no-log-timestamps --no-jinja --chat-te
mplate chatml --load-mode none --flash-attn auto -b 512 -ub 512 --context-shift --keep 4
ollama    6514   0.4   0.6 2614576 39692 ?        Ssl  12:14   1:34 /usr/local/bin/ollama serve
root      8053   0.2   0.4 1810992 29312 pts/4    Sl+  17:38   0:00 ollama run Asistente
root      8025   0.0   0.1 14332   6996 pts/3    S+   17:37   0:00 sudo su
root      8005   0.0   0.0 3080    1024 ?        S    17:37   0:00 /init
root       1   0.0   0.2 22480   13160 ?        Ss   12:04   0:04 /sbin/init
root      8028   0.0   0.0 5812    4736 pts/4    S    17:37   0:00 bash
jbara     8012   0.0   0.0 6076    5248 pts/3    Ss   17:37   0:00 -bash
root      621   0.0   0.1 14328   7000 pts/0    S+   12:04   0:01 sudo su
root@BOG30487:/mnt/c/Users/jbara# ps aux --sort=-%mem | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
ollama    8060  62.9  24.7 2136124 1504716 ?        Sl   17:38   1:34 /usr/local/lib/ollama/llama-server --model /usr/share
/ollama/.ollama/models/blobs/sha256-74701a8c35f6c8d9a4b91f3f3497643001d63e0c7a84e085bed452548fa88d45 --port 37815 --host
127.0.0.1 --no-webui --offline -c 4096 -np 1 --log-verbosity 4 --no-log-prefix --no-log-timestamps --no-jinja --chat-te
mplate chatml --load-mode none --flash-attn auto -b 512 -ub 512 --context-shift --keep 4
ollama    6514   0.4   0.6 2614576 39692 ?        Ssl  12:14   1:34 /usr/local/bin/ollama serve
root      8053   0.2   0.4 1810992 29312 pts/4    Sl+  17:38   0:00 ollama run Asistente
root      247   0.0   0.3 107028 22268 ?        Ssl  12:04   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unatt
ended-upgrade-shutdown --wait-for-signal
root      42   0.0   0.2 66900   15740 ?        S<s  12:04   0:00 /usr/lib/systemd/systemd-journald
root     191   0.0   0.2 68744   14720 ?        Ss   12:04   0:00 /usr/sbin/virtlogd
root     188   0.0   0.2 68740   14592 ?        Ss   12:04   0:00 /usr/sbin/virtlockd
root       1   0.0   0.2 22480   13160 ?        Ss   12:04   0:04 /sbin/init
```

Ejercicio 4 – Administración del servicio Ollama

```
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# systemctl status ollama --no-pager
● ollama.service - Ollama Service
   Loaded: loaded (/etc/systemd/system/ollama.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-30 10:48:26 -05; 6h ago
     Main PID: 6514 (ollama)
       Tasks: 29 (Limit: 7100)
      Memory: 2.6G (peak: 2.7G)
         CPU: 3min 17.148s
    CGroup: /system.slice/ollama.service
            └─6514 /usr/local/bin/ollama serve
               └─8060 /usr/local/lib/ollama/llama-server --model /usr/share/ollama/.ollama/models/blobs/sha256-74701a8c35...

Sep 30 17:39:35 BOG30487 ollama[6514]: slot print_timing: id 0 | task 0 | n_gen = 648, tg = 18.27 t/s, tg_...7.82 t/s
Sep 30 17:39:38 BOG30487 ollama[6514]: slot print_timing: id 0 | task 0 | n_gen = 702, tg = 18.24 t/s, tg_...7.86 t/s
Sep 30 17:39:41 BOG30487 ollama[6514]: slot print_timing: id 0 | task 0 | n_gen = 753, tg = 18.13 t/s, tg_...6.68 t/s
Sep 30 17:39:44 BOG30487 ollama[6514]: slot print_timing: id 0 | task 0 | prompt eval time = 855.73 ms / ... second)
Sep 30 17:39:44 BOG30487 ollama[6514]: slot print_timing: id 0 | task 0 | eval time = 43987.72 ms / ... second)
Sep 30 17:39:44 BOG30487 ollama[6514]: slot print_timing: id 0 | task 0 | total time = 44843.44 ms / ...7 tokens
Sep 30 17:39:44 BOG30487 ollama[6514]: slot print_timing: id 0 | task 0 | graphs reused = 792
Sep 30 17:39:44 BOG30487 ollama[6514]: slot release: id 0 | task 0 | stop processing: n_tokens = 856, tru...ated = 0
Sep 30 17:39:44 BOG30487 ollama[6514]: srv update_slots: all slots are idle
Sep 30 17:39:44 BOG30487 ollama[6514]: [GIN] 2026/09/30 - 17:39:44 | 200 | 44.856925725s | 127.0.0.1 | PO...pi/chat"
Hint: Some lines were ellipsized, use -l to show in full.
root@BOG30487:/mnt/c/Users/jbara#
```

```
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# systemctl stop ollama
root@BOG30487:/mnt/c/Users/jbara# curl http://127.0.0.1:11434/api/tags
curl: (7) Failed to connect to 127.0.0.1 port 11434 after 0 ms: Couldn't connect to server
root@BOG30487:/mnt/c/Users/jbara# systemctl start ollama
root@BOG30487:/mnt/c/Users/jbara# curl http://127.0.0.1:11434/api/tags
{"models":[{"name":"asistente-ciberseguridad:latest","model":"asistente-ciberseguridad:latest","modified_at":"2026-09-30T12:17:12.580598233-05:00","size":1321098262,"digest":"d6e4fe34ec6782bc5eb018cfea479382203fcb50baea948d708d60a2b8056cab","details":{"parent_model":"llama3.2:1b","format":"gguf","family":"llama","families":["llama"],"parameter_size":"1.2B","quantization_level":"Q8_0","context_length":131072,"embedding_length":2048},"capabilities":{"completion","tools"}}],"name":"Asistente:latest","model":"Asistente:latest","modified_at":"2026-09-30T11:15:00.070218657-05:00","size":1321098329,"digest":"baf6a787fdffd633537aa2eb51cfd54cb93ff08e28040095462bb63daf552878","details":{"parent_model":"","format":"gguf","family":"llama","families":["llama"],"parameter_size":"1.2B","quantization_level":"Q8_0","context_length":131072,"embedding_length":2048},"capabilities":{"completion","tools"}}]}root@BOG30487:/mnt/c/Users/jbara# journalctl -u ollama -n 50 --no-pager
Sep 30 17:38:59 BOG30487 ollama[6514]: slot get_availabl: id 0 | task -1 | selected slot by LRU, t_last = -1
Sep 30 17:38:59 BOG30487 ollama[6514]: srv get_availabl: updating prompt cache
Sep 30 17:38:59 BOG30487 ollama[6514]: srv load: - looking for better prompt, base f_keep = -1.000, f_sim = 0.000
Sep 30 17:38:59 BOG30487 ollama[6514]: srv update: - cache state: 0 prompts, 0.000 MiB (limits: 8192.000 MiB, 4096 tokens, 8589934592 est)
Sep 30 17:38:59 BOG30487 ollama[6514]: srv get_availabl: prompt cache update took 0.01 ms
Sep 30 17:38:59 BOG30487 ollama[6514]: slot launch_slot: id 0 | task -1 | sampler chain: logits -> ?penalties -> ?dry -> ?top-n-sigma -> top-k -> ?typical -> top-p -> ?min-p -> ?xtc -> temp-ext -> dist
Sep 30 17:38:59 BOG30487 ollama[6514]: slot launch_slot: id 0 | task -1 | sampler params:
Sep 30 17:38:59 BOG30487 ollama[6514]: repeat_last_n = 64, repeat_penalty = 1.000, frequency_penalty = 0.000, pr
```

Ejercicio 5 – Procesos, señales y servicios.

```
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# pgrep -a ollama
8053 ollama run Asistente
8095 /usr/local/bin/ollama serve
root@BOG30487:/mnt/c/Users/jbara# ps -fp 8053
UID      PID     PPID  C  STIME TTY          TIME CMD
root      8053      0 19:23 pts/4    00:00:00 ollama run Asistente
root@BOG30487:/mnt/c/Users/jbara# ps -fp 8095
UID      PID     PPID  C  STIME TTY          TIME CMD
ollama    8095       1  0 19:28 ?        00:00:00 /usr/local/bin/ollama serve
root@BOG30487:/mnt/c/Users/jbara# systemctl status ollama
● ollama.service - Ollama Service
   Loaded: loaded (/etc/systemd/system/ollama.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-30 17:42:55 -05; 4h 48min ago
     Main PID: 8095 (ollama)
       Tasks: 10 (limit: 7100)
      Memory: 10.7M (peak: 124.4M)
         CPU: 726ms
        CGroup: /system.slice/ollama.service
                └─8095 /usr/local/bin/ollama serve

Sep 30 17:42:55 BOG30487 ollama[8095]: time=2026-09-30T17:42:55.153-05:00 level=INFO source=routes.go:2099 msg="server >
Sep 30 17:42:55 BOG30487 ollama[8095]: time=2026-09-30T17:42:55.153-05:00 level=INFO source=routes.go:2101 msg="Ollama >
Sep 30 17:42:55 BOG30487 ollama[8095]: time=2026-09-30T17:42:55.155-05:00 level=INFO source=images.go:922 msg="total bl >
Sep 30 17:42:55 BOG30487 ollama[8095]: time=2026-09-30T17:42:55.155-05:00 level=INFO source=images.go:930 msg="total un >
Sep 30 17:42:55 BOG30487 ollama[8095]: time=2026-09-30T17:42:55.156-05:00 level=INFO source=routes.go:2159 msg="Listen >
Sep 30 17:42:55 BOG30487 ollama[8095]: time=2026-09-30T17:42:55.159-05:00 level=INFO source=runner.go:60 msg="discoveri >
Sep 30 17:42:55 BOG30487 ollama[8095]: time=2026-09-30T17:42:55.494-05:00 level=INFO source=model_recommendations.go:17 >
Sep 30 17:42:55 BOG30487 ollama[8095]: time=2026-09-30T17:42:55.778-05:00 level=INFO source=types.go:50 msg="inference >
```

Ejercicio 6 – Red y Puerto de Ollama

```
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# ss -lntp | grep 11434
LISTEN 0      4096      127.0.0.1:11434      0.0.0.0:*      users:((("ollama",pid=8095,fd=3))
root@BOG30487:/mnt/c/Users/jbara# curl http://127.0.0.1:11434/api/tags
{"models":[{"name":"asistente-ciberseguridad:latest","model":"asistente-ciberseguridad:latest","modified_at":"2026-09-30
T12:17:12.580598233-05:00","size":1321098262,"digest":"d6e4fe34ec6782bc5eb018cfea479382203fcb50bae948d708d60a2b8056cab"
,"details":{"parent_model":"llama3.2:1b","format":"gguf","family":"llama","families":["llama"],"parameter_size":"1.2B","
quantization_level":"Q8_0","context_length":131072,"embedding_length":2048},"capabilities":{"completion","tools"}},{name
e":"Asistente:latest","model":"Asistente:latest","modified_at":"2026-09-30T11:15:00.070218657-05:00","size":1321098329,"
digest":"baf6a787fdffd633537aa2eb51cfd54cb93ff08e28040095462bb63daf552878","details":{"parent_model":"","format":"gguf",
"family":"llama","families":["llama"],"parameter_size":"1.2B","quantization_level":"Q8_0","context_length":131072,"embed
ding_length":2048},"capabilities":{"completion","tools"}},{name":"llama3.2:1b","model":"llama3.2:1b","modified_at":"202
6-09-30T11:04:50.693824384-05:00","size":1321098329,"digest":"baf6a787fdffd633537aa2eb51cfd54cb93ff08e28040095462bb63daf
552878","details":{"parent_model":"","format":"gguf","family":"llama","families":["llama"],"parameter_size":"1.2B","quan
tization_level":"Q8_0","context_length":131072,"embedding_length":2048},"capabilities":{"completion","tools"}}]}root@BOG
30487:/mnt/c/Users/jbara# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:5d:1a:3f brd ff:ff:ff:ff:ff:ff
    inet 172.20.14.233/20 brd 172.20.15.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe5d:1a3f/64 scope link
```

Ejercicio 7 – Almacenamiento de modelos

```
root@BOG30487:/mnt/c/Use x root@BOG30487:/mnt/c/User x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# df -h
Filesystem      Size  Used Avail Use% Mounted on
none            2.9G   0    2.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            2.9G  4.0K   2.9G   1% /mnt/wsl
drivers          476G  279G  198G   59% /usr/lib/wsl/drivers
/dev/sdd        1007G   11G  945G   2% /
none            2.9G  120K   2.9G   1% /mnt/wslg
none            2.9G   0    2.9G   0% /usr/lib/wsl/lib
rootfs           2.9G  2.7M   2.9G   1% /init
none            2.9G  556K   2.9G   1% /run
none            2.9G   0    2.9G   0% /run/lock
none            2.9G   0    2.9G   0% /run/shm
none            2.9G   76K   2.9G   1% /mnt/wslg/versions.txt
none            2.9G   76K   2.9G   1% /mnt/wslg/doc
C:\              476G  279G  198G   59% /mnt/c
tmpfs            2.9G  220K   2.9G   1% /run/qemu
tmpfs            2.9G   16K   2.9G   1% /run/user/1000
tmpfs            593M   16K   593M   1% /run/user/0
root@BOG30487:/mnt/c/Users/jbara#
```

```
root@BOG30487:/mnt/c/Use x root@BOG30487:/mnt/c/User x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# ollama pull deepseek-r1:1.5b
pulling manifest
pulling aabd4debf0c8: 100% | 1.1 GB/1.1 GB 8.8 MB/s 0s
verifying sha256 digest
writing manifest
success
root@BOG30487:/mnt/c/Users/jbara# df -h
Filesystem      Size  Used Avail Use% Mounted on
none            2.9G   0    2.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            2.9G  4.0K   2.9G   1% /mnt/wsl
drivers          476G  280G  197G   59% /usr/lib/wsl/drivers
/dev/sdd        1007G   12G  944G   2% /
none            2.9G  120K   2.9G   1% /mnt/wslg
none            2.9G   0    2.9G   0% /usr/lib/wsl/lib
rootfs           2.9G  2.7M   2.9G   1% /init
none            2.9G  556K   2.9G   1% /run
none            2.9G   0    2.9G   0% /run/lock
none            2.9G   0    2.9G   0% /run/shm
none            2.9G   76K   2.9G   1% /mnt/wslg/versions.txt
none            2.9G   76K   2.9G   1% /mnt/wslg/doc
C:\              476G  280G  197G   59% /mnt/c
tmpfs            2.9G  220K   2.9G   1% /run/qemu
tmpfs            2.9G   16K   2.9G   1% /run/user/1000
tmpfs            593M   16K   593M   1% /run/user/0
root@BOG30487:/mnt/c/Users/jbara#
```

Ejercicio 8 – Rendimiento y selección del modelo.

Ejercicio 9 – Automatización con Bash.

```
root@BOG30487: /mnt/c/Use x root@BOG30487: /mnt/c/User x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# cat > reporte_sistema.sh
#!/bin/bash
echo "=== REPORTE DEL SISTEMA ==="
date
hostname
uname -a
echo "---- CPU ----"
nproc
lscpu | grep "Model name"
echo "---- RAM ----"
free -h
echo "---- DISCO ----"
df -h /
echo "---- OLLAMA ----"
systemctl is-active ollama^[[A^[[A^[[B^[[B^Z
[1]+  Stopped cat > reporte_sistema.sh
root@BOG30487:/mnt/c/Users/jbara# cat > reporte_sistema.sh
#!/bin/bash
echo "=== REPORTE DEL SISTEMA ==="
date
hostname
uname -a
echo "---- CPU ----"
nproc
lscpu | grep "Model name"
echo "---- RAM ----"
free -h
echo "---- DISCO ----"
```

```
root@BOG30487: /mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# echo Johan David Baracaldo Caicedo
Johan David Baracaldo Caicedo
root@BOG30487:/mnt/c/Users/jbara# ./reporte_sistema.sh | tee reporte.txt
=== REPORTE DEL SISTEMA ===
Wed Sep 30 22:50:57 -05 2026
BOG30487
Linux BOG30487 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46 UTC 2025 x86_64 x86_64 x86_64
GNU/Linux
--- CPU ---
4
--- RAM ---
Mem:          total      used      free      shared buff/cache available
Swap:         2.0Gi       0B       2.0Gi       3.8Mi    281Mi    5.3Gi
--- DISCO ---
Filesystem    Size  Used Avail Use% Mounted on
/dev/sdd      1007G  12G  944G   2% /
--- OLLAMA ---
active
root@BOG30487:/mnt/c/Users/jbara#
```

Ejercicio 10 – IA como herramienta de apoyo del sistema operativo

```
root@BOG30487:/mnt/c/Use x + v
root@BOG30487:/mnt/c/Users/jbara# free -h
              total        used        free      shared  buff/cache   available
Mem:          5.8Gi        529Mi        5.2Gi        3.8Mi        281Mi        5.3Gi
Swap:         2.0Gi           0B        2.0Gi
root@BOG30487:/mnt/c/Users/jbara# df -h
Filesystem      Size  Used Avail Use% Mounted on
none            2.9G   0 2.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            2.9G  4.0K 2.9G   1% /mnt/wsl
drivers         476G  280G 197G  59% /usr/lib/wsl/drivers
/dev/sdd       1007G   12G 944G   2% /
none            2.9G 120K 2.9G   1% /mnt/wslg
none            2.9G   0 2.9G   0% /usr/lib/wsl/lib
rootfs          2.9G  2.7M 2.9G   1% /init
none            2.9G 552K 2.9G   1% /run
none            2.9G   0 2.9G   0% /run/lock
none            2.9G   0 2.9G   0% /run/shm
none            2.9G  76K 2.9G   1% /mnt/wslg/versions.txt
none            2.9G  76K 2.9G   1% /mnt/wslg/doc
C:\             476G  280G 197G  59% /mnt/c
tmpfs           2.9G  220K 2.9G   1% /run/qemu
tmpfs           2.9G   16K 2.9G   1% /run/user/1000
tmpfs           593M   16K 593M   1% /run/user/0
root@BOG30487:/mnt/c/Users/jbara# systemctl status ollama
● ollama.service - Ollama Service
   Loaded: loaded (/etc/systemd/system/ollama.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-30 17:42:55 -05; 5h 9min ago
     Main PID: 8095 (ollama)
       Tasks: 16 (limit: 7100)
      Memory: 17.9M (peak: 1.0G)
         CPU: 43.091s
```

root@BOG30487:/mnt/c/Users/jbara# free -h

	total	used	free	shared	buff/cache	available
Mem:	5.8Gi	529Mi	5.2Gi	3.8Mi	281Mi	5.3Gi
Swap:	2.0Gi	0B	2.0Gi			

root@BOG30487:/mnt/c/Users/jbara# df -h

Filesystem	Size	Used	Avail	Use%	Mounted on
none	2.9G	0	2.9G	0%	/usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none	2.9G	4.0K	2.9G	1%	/mnt/wsl
drivers	476G	280G	197G	59%	/usr/lib/wsl/drivers
/dev/sdd	1007G	12G	944G	2%	/
none	2.9G	120K	2.9G	1%	/mnt/wslg
none	2.9G	0	2.9G	0%	/usr/lib/wsl/lib
rootfs	2.9G	2.7M	2.9G	1%	/init
none	2.9G	552K	2.9G	1%	/run
none	2.9G	0	2.9G	0%	/run/lock
none	2.9G	0	2.9G	0%	/run/shm

none 2.9G 76K 2.9G 1% /mnt/wslg/versions.txt

none 2.9G 76K 2.9G 1% /mnt/wslg/doc

C:\ 476G 280G 197G 59% /mnt/c

tmpfs 2.9G 220K 2.9G 1% /run/qemu

tmpfs 2.9G 16K 2.9G 1% /run/user/1000

tmpfs 593M 16K 593M 1% /run/user/0

root@BOG30487:/mnt/c/Users/jbara# systemctl status ollama

● ollama.service - Ollama Service

Loaded: loaded (/etc/systemd/system/ollama.service; enabled; preset: enabled)

Active: active (running) since Wed 2026-09-30 17:42:55 -05; 5h 9min ago

Main PID: 8095 (ollama)

Tasks: 16 (limit: 7100)

Memory: 17.9M (peak: 1.0G)

CPU: 43.091s

CGroup: /system.slice/ollama.service

└─8095 /usr/local/bin/ollama serve

Sep 30 22:33:27 BOG30487 ollama[8095]: [GIN] 2026/09/30 - 22:33:27 | 200 | 19.496594ms | 127.0.0.1 | GET ">

Sep 30 22:36:30 BOG30487 ollama[8095]: [GIN] 2026/09/30 - 22:36:30 | 200 | 1.590911ms | 127.0.0.1 | HEAD ">

Sep 30 22:36:31 BOG30487 ollama[8095]: time=2026-09-30T22:36:31.445-05:00 level=INFO source=download.go:182 msg="downlo>

Sep 30 22:38:42 BOG30487 ollama[8095]: time=2026-09-30T22:38:42.288-05:00 level=INFO source=download.go:182 msg="downlo>

Sep 30 22:38:42 BOG30487 ollama[8095]: time=2026-09-30T22:38:42.900-05:00 level=INFO source=download.go:182 msg="downlo>

Sep 30 22:38:43 BOG30487 ollama[8095]: time=2026-09-30T22:38:43.425-05:00
level=INFO source=download.go:182 msg="downlo>

Sep 30 22:38:43 BOG30487 ollama[8095]: time=2026-09-30T22:38:43.995-05:00
level=INFO source=download.go:182 msg="downlo>

Sep 30 22:38:45 BOG30487 ollama[8095]: [GIN] 2026/09/30 - 22:38:45 | 200 |
2m14s | 127.0.0.1 | POST ">

Sep 30 22:46:34 BOG30487 ollama[8095]: [GIN] 2026/09/30 - 22:46:34 | 200 |
52.816µs | 127.0.0.1 | HEAD ">

Sep 30 22:46:34 BOG30487 ollama[8095]: [GIN] 2026/09/30 - 22:46:34 | 200 |
71.013847ms | 127.0.0.1 | GET ">

lines 1-20/20 (END)

^C

root@BOG30487:/mnt/c/Users/jbara# ^C

root@BOG30487:/mnt/c/Users/jbara# ps aux --sort=%mem | head

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	453	0.0	0.0	3064	896	?	Ss	13:49	0:00	/init
root	457	0.0	0.0	3080	1024	?	S	13:49	0:01	/init
root	424	0.0	0.0	11492	1068	?	S	13:49	0:00	/usr/sbin/dnsmasq --conf- file=/var/lib/libvirt/dnsmasq/default.conf --leasefile-ro --dhcp- script=/usr/lib/libvirt/libvirt_leaseshelper
root	253	0.0	0.0	3120	1664	tty1	Ss+	13:49	0:00	/sbin/agetty -o -p -- \u -- noclear - linux
root	2	0.0	0.0	3060	1792	?	Sl	13:49	0:00	/init
root	6	0.0	0.0	3060	1792	?	Sl	13:49	0:00	plan9 --control-socket 7 --log- level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root	232	0.0	0.0	3164	1920	hvc0	Ss+	13:49	0:00	/sbin/agetty -o -p -- \u -- noclear --keep-baud - 115200,38400,9600 vt220
root	165	0.0	0.0	4240	2432	?	Ss	13:49	0:00	/usr/sbin/cron -f -P
root	622	0.0	0.0	14328	2540	pts/2	Ss	13:50	0:00	sudo su

```
root@BOG30487:/mnt/c/Users/jbara# ollama run Asistente
```

```
>>> Interpreta estos datos: root@BOG30487:/mnt/c/Users/jbara# free -h
```

```
...      total    used    free   shared  buff/cache   available
... Mem:    5.8Gi   529Mi   5.2Gi    3.8Mi    281Mi    5.3Gi
... Swap:    2.0Gi     0B    2.0Gi
```

```
... root@BOG30487:/mnt/c/Users/jbara# df -h
```

```
... Filesystem    Size  Used Avail Use% Mounted on
... none          2.9G   0 2.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
... none          2.9G 4.0K 2.9G   1% /mnt/wsl
... drivers       476G 280G 197G  59% /usr/lib/wsl/drivers
... /dev/sdd      1007G 12G 944G   2% /
... none          2.9G 120K 2.9G   1% /mnt/wslg
... none          2.9G   0 2.9G   0% /usr/lib/wsl/lib
... rootfs        2.9G 2.7M 2.9G   1% /init
... none          2.9G 552K 2.9G   1% /run
... none          2.9G   0 2.9G   0% /run/lock
... none          2.9G   0 2.9G   0% /run/shm
... none          2.9G 76K 2.9G   1% /mnt/wslg/versions.txt
... none          2.9G 76K 2.9G   1% /mnt/wslg/doc
... C:\           476G 280G 197G  59% /mnt/c
... tmpfs         2.9G 220K 2.9G   1% /run/qemu
... tmpfs         2.9G 16K 2.9G   1% /run/user/1000
... tmpfs         593M 16K 593M   1% /run/user/0
```

```
... root@BOG30487:/mnt/c/Users/jbara# systemctl status ollama
```

```
... ● ollama.service - Ollama Service
```

```
...    Loaded: loaded (/etc/systemd/system/ollama.service; enabled; preset: enabled)
```

```
... Active: active (running) since Wed 2026-09-30 17:42:55 -05; 5h 9min ago
... Main PID: 8095 (ollama)
... Tasks: 16 (limit: 7100)
... Memory: 17.9M (peak: 1.0G)
... CPU: 43.091s
... CGroup: /system.slice/ollama.service
...     └─8095 /usr/local/bin/ollama serve
...
... Sep 30 22:33:27 BOG30487 ollama[8095]: [GIN] 2026/09/30 - 22:33:27 | 200 |
19.496594ms | 127.0.0.1 | GET
... ">
... Sep 30 22:36:30 BOG30487 ollama[8095]: [GIN] 2026/09/30 - 22:36:30 | 200 |
1.590911ms | 127.0.0.1 | HEAD
... ">
... Sep 30 22:36:31 BOG30487 ollama[8095]: time=2026-09-30T22:36:31.445-05:00
level=INFO source=download.go:182 msg="dow
... nlo>
... Sep 30 22:38:42 BOG30487 ollama[8095]: time=2026-09-30T22:38:42.288-05:00
level=INFO source=download.go:182 msg="dow
... nlo>
... Sep 30 22:38:42 BOG30487 ollama[8095]: time=2026-09-30T22:38:42.900-05:00
level=INFO source=download.go:182 msg="dow
... nlo>
... Sep 30 22:38:43 BOG30487 ollama[8095]: time=2026-09-30T22:38:43.425-05:00
level=INFO source=download.go:182 msg="dow
... nlo>
... Sep 30 22:38:43 BOG30487 ollama[8095]: time=2026-09-30T22:38:43.995-05:00
level=INFO source=download.go:182 msg="dow
```

```

... nlo>

... Sep 30 22:38:45 BOG30487 ollama[8095]: [GIN] 2026/09/30 - 22:38:45 | 200 |
2m14s | 127.0.0.1 | POST

... ">

... Sep 30 22:46:34 BOG30487 ollama[8095]: [GIN] 2026/09/30 - 22:46:34 | 200 |
52.816µs | 127.0.0.1 | HEAD

... ">

... Sep 30 22:46:34 BOG30487 ollama[8095]: [GIN] 2026/09/30 - 22:46:34 | 200 |
71.013847ms | 127.0.0.1 | GET

... ">

... lines 1-20/20 (END)

... ^C

... root@BOG30487:/mnt/c/Users/jbara# ^C

... root@BOG30487:/mnt/c/Users/jbara# ps aux --sort=%mem | head

... USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
... root      453  0.0  0.0  3064  896 ?        Ss   13:49   0:00 /init
... root      457  0.0  0.0  3080 1024 ?        S    13:49   0:01 /init
... root      424  0.0  0.0 11492 1068 ?        S    13:49   0:00 /usr/sbin/dnsmasq --conf-
file=/var/lib/libvirt/dn
... smasq/default.conf --leasefile-ro --dhcp-script=/usr/lib/libvirt/libvirt_leaseshelper
... root      253  0.0  0.0  3120 1664 tty1    Ss+  13:49   0:00 /sbin/agetty -o -p -- \u --
noclear - linux
... root        2  0.0  0.0  3060 1792 ?        Sl   13:49   0:00 /init
... root        6  0.0  0.0  3060 1792 ?        Sl   13:49   0:00 plan9 --control-socket 7 --log-
level 4 --server-f
... d 8 --pipe-fd 10 --log-truncate
... root      232  0.0  0.0  3164 1920 hvc0    Ss+  13:49   0:00 /sbin/agetty -o -p -- \u --
noclear --keep-baud -

```

... 115200,38400,9600 vt220

... root 165 0.0 0.0 4240 2432 ? Ss 13:49 0:00 /usr/sbin/cron -f -P

... root 622 0.0 0.0 14328 2540 pts/2 Ss 13:50 0:00 sudo su

Aquí te presento un resumen de los datos que se han proporcionado:

* El sistema operativo que se está ejecutando es Windows 10, específicamente el sistema de distribución WSL

(Windows Subsystem for Linux).

* El sistema de archivos total se encuentra en 5,8 GB, con 529 MiB de espacio utilizado y 5,2 GB de espacio libre.

* El espacio disponible en el sistema de archivos total se encuentra en 3,8 GB.

* El espacio utilizado por el sistema de archivos total se encuentra en 529 MiB.

* El espacio disponible en el sistema de archivos total se encuentra en 5,3 GB.

* El espacio utilizado por el sistema de archivos total se encuentra en 2,0 GB.

* El sistema de archivos mounted en `/usr/lib/modules/6.6.87.2-microsoft-standard-WSL2` se encuentra en 2,9 GiB de espacio libre.

* El sistema de archivos mounted en `/mnt/wsl` se encuentra en 2,9 GiB de espacio libre.

* El sistema de archivos mounted en `/run/wsl` se encuentra en 2,9 GiB de espacio libre.

* El sistema de archivos mounted en `/mnt/wslg` se encuentra en 2,9 GiB de espacio libre.

* El sistema de archivos mounted en `/run/lock` se encuentra en 2,9 GiB de espacio libre.

* El sistema de archivos mounted en `/run/shm` se encuentra en 2,9 GiB de espacio libre.

* El sistema de archivos mounted en `/mnt/wslg/versions.txt` se encuentra en 2,9 GiB de espacio libre.

- * El sistema de archivos mounted en `/mnt/wslg/doc` se encuentra en 2,9 GiB de espacio libre.
- * El sistema de archivos mounted en `/C:` se encuentra en 476 GiB de espacio libre.
- * El sistema de archivos `/tmpfs` se encuentra en 2,9 GiB de espacio libre.
- * El sistema de archivos `/run/qemu` se encuentra en 593 MiB de espacio libre.
- * El sistema de archivos `/run/user/1000` se encuentra en 593 MiB de espacio libre.
- * El sistema de archivos `/run/user/0` se encuentra en 593 MiB de espacio libre.
- * El sistema de archivos `/dev/sdd` se encuentra en 1007 GiB de espacio libre.
- * El sistema de archivos `/none` se encuentra en 2,9 GiB de espacio libre.
- * El sistema de archivos `/mnt/wsl` se encuentra en 2,9 GiB de espacio libre.
- * El sistema de archivos `/run/wsl` se encuentra en 2,9 GiB de espacio libre.
- * El sistema de archivos `/run/lock` se encuentra en 2,9 GiB de espacio libre.
- * El sistema de archivos `/run/shm` se encuentra en 2,9 GiB de espacio libre.
- * El sistema de archivos `/mnt/wslg/versions.txt` se encuentra en 2,9 GiB de espacio libre.
- * El sistema de archivos `/mnt/wslg/doc` se encuentra en 2,9 GiB de espacio libre.

En cuanto a la sesión de usuario, se puede ver que el usuario `root` tiene una historia de comandos con una lista

de los servicios y comandos que se han ejecutado.

En cuanto a la ejecución de comandos, se puede ver que el usuario `root` ha ejecutado varios comandos, incluyendo

`ps aux`, `df -h`, `systemctl status ollama`, y `ps aux`.